

On the Universality of Comparability Grids for Measurement-Based Quantum Computation

Ryan Y. Batubara

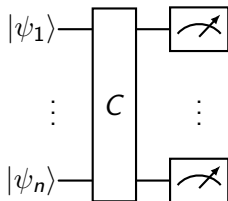
Advisor: Professor Daniel Grier

UCSD

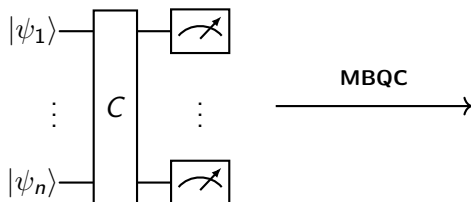
Mathematics Undergraduate Honors Presentation

February 13, 2026

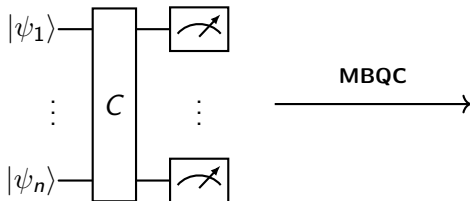
Measurement-Based Quantum Computation (MBQC)



Measurement-Based Quantum Computation (MBQC)

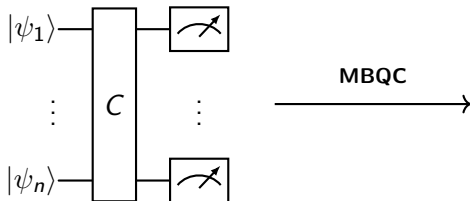


Measurement-Based Quantum Computation (MBQC)



- ▶ Single-qubit measurements on a
- ▶ "resource state"

Measurement-Based Quantum Computation (MBQC)



- ▶ Single-qubit measurements on a
- ▶ "resource state"
- ▶ Excluding resource state, done in constant depth.

Example Resource State: Graph State

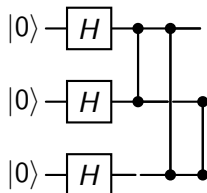
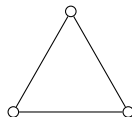
Let $G = (V, E)$ be a graph. The **graph state** $|G\rangle$ has

- ▶ For every $v \in V$, prepare a qubit $|+\rangle = H|0\rangle$.
- ▶ For every $e = (u, v) \in E$, apply CZ on qubits of u and v .

Example Resource State: Graph State

Let $G = (V, E)$ be a graph. The **graph state** $|G\rangle$ has

- ▶ For every $v \in V$, prepare a qubit $|+\rangle = H|0\rangle$.
- ▶ For every $e = (u, v) \in E$, apply CZ on qubits of u and v .



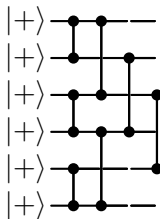
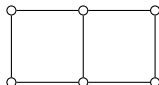
Grid Graph State

Let $\text{Grid}_{m \times n}$ be the grid graph with m rows and n columns.
Let $|\text{Grid}_{m \times n}\rangle$ be its graph state.

Grid Graph State

Let $\text{Grid}_{m \times n}$ be the grid graph with m rows and n columns.
Let $|\text{Grid}_{m \times n}\rangle$ be its graph state.

Example: $|\text{Grid}_{2 \times 3}\rangle$.



Universal Resource, Informally

Definition

Let LO denote local operations, i.e. single qubit unitaries and measurements. Let CC denote classical communication.

Universal Resource, Informally

Definition

Let LO denote local operations, i.e. single qubit unitaries and measurements. Let CC denote classical communication.

Definition

Let Ψ be a family of resource states. Ψ is a **universal resource** for MBQC if for every $U \in SU(2^n)$, $\forall n \in \mathbb{N}$, there is a sequence of LOCC and some resource state $|\psi\rangle \in \Psi$, such that

$$|\psi\rangle \xrightarrow{\text{LOCC}} U|0^n\rangle$$

Universal Resource, Informally

Definition

Let LO denote local operations, i.e. single qubit unitaries and measurements. Let CC denote classical communication.

Definition

Let Ψ be a family of resource states. Ψ is a **universal resource** for MBQC if for every $U \in SU(2^n)$, $\forall n \in \mathbb{N}$, there is a sequence of LOCC and some resource state $|\psi\rangle \in \Psi$, such that

$$|\psi\rangle \xrightarrow{\text{LOCC}} U|0^n\rangle$$

In other words, we can “simulate any state” using LOCC and a resource state from Ψ .

Grid is a Universal Resource

Theorem

$\{\text{Grid}_{1 \times i}\}_{i \in \mathbb{N}}$ (The lines) are a 1-qubit universal resource for MBQC.

Lemma (Euler decomposition)

Every single-qubit unitary can be expressed (up to global phase) as the product

$$R_z(\theta_3)R_x(\theta_2)R_z(\theta_1)$$

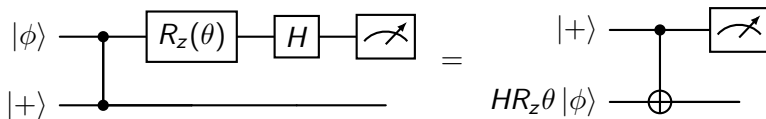
for some $\theta_1, \theta_2, \theta_3 \in [0, 2\pi]$.

Lemma

$R_x(\theta) = HR_z(\theta)H$, for all $\theta \in [0, 2\pi]$.

Grid is a Universal Resource cont'd

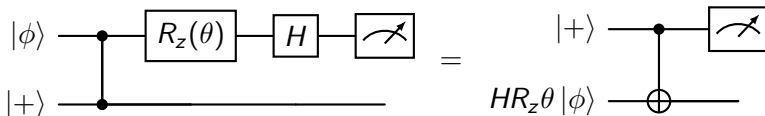
Informal proof.



□

Grid is a Universal Resource cont'd

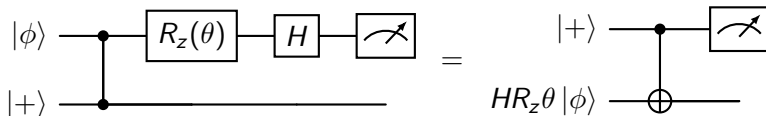
Informal proof.



If we measure 0, we get $HR_z(\theta)|\phi\rangle$ in the second register. If we measure 1, we get $XH_z(\theta)|\phi\rangle$. Thus given the right measurements we can apply any 1-qubit gate. □

Grid is a Universal Resource cont'd

Informal proof.



If we measure 0, we get $HR_z(\theta)|\phi\rangle$ in the second register. If we measure 1, we get $XH_z(\theta)|\phi\rangle$. Thus given the right measurements we can apply any 1-qubit gate. □

Theorem

Let Grid denote the family of all grid graphs. Grid is a universal resource for MBQC.

Comparability Grid

Definition

Let $\text{CGrid}_{m \times n}$ denote the m rows, n column **comparability grid**, which is an undirected graph $G = (V, E)$ with

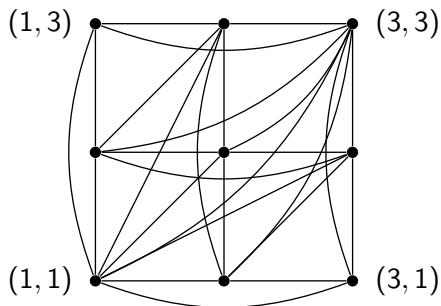
$$V = \{(i, j) \mid i \in [m], j \in [n]\}$$

$$E = \{\{(i_1, j_1), (i_2, j_2)\} \mid (i_1, j_1), (i_2, j_2) \in V, i_2 \geq i_1, j_2 \geq j_1.\}$$

i.e. the grid with partial order $\leq$ defining edges.

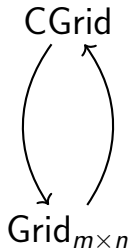
Comparability Grid Example

Example: $CGrid_{3 \times 3}$:



Proving Comparability Grid Universality

Strategy: Embed Grid in CGrid. Two ways to do this:

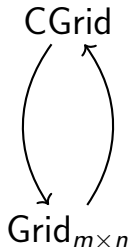


Proving Comparability Grid Universality

Strategy: Embed Grid in CGrid. Two ways to do this:

Method 1

What grids can we get
from vertex deletion?

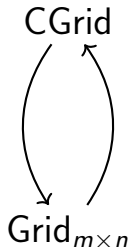


Proving Comparability Grid Universality

Strategy: Embed Grid in CGrid. Two ways to do this:

Method 1

What grids can we get
from vertex deletion?



Method 2

How can we embed
Grid_{m × n} in CGrid?

$\text{Grid}_{2 \times n}$ is a subgraph

Lemma

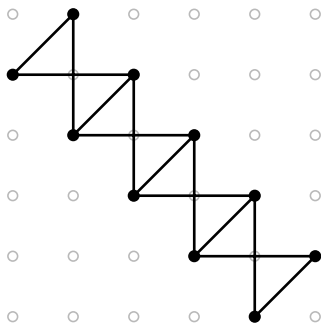
For all $n \in \mathbb{N}$, $\text{Grid}_{2 \times n}$ is a subgraph of a sufficiently large CGrid.

$\text{Grid}_{2 \times n}$ is a subgraph

Lemma

For all $n \in \mathbb{N}$, $\text{Grid}_{2 \times n}$ is a subgraph of a sufficiently large CGrid.

Proof.



Which grids are subgraphs?

Theorem

Grid $_{k \times n}$ is not a subgraph of any CGrid for $k > 2$, $n > 2$.

Which grids are subgraphs?

Theorem

Grid $_{k \times n}$ is not a subgraph of any CGrid for $k > 2$, $n > 2$.

Proof.

Using a computer, we enumerate all possible 9 vertex subgraphs of CGrid. No such subgraph is isomorphic to Grid $_{3 \times 3}$. □

Permutation Representation of Subgraphs

Lemma

Any subgraph H of $CGrid$ with $|V(H)| = n$ can be uniquely identified as a permutation of $[n]$, denoted $\pi(H)$.

Permutation Representation of Subgraphs

Lemma

Any subgraph H of CGrid with $|V(H)| = n$ can be uniquely identified as a permutation of $[n]$, denoted $\pi(H)$.

Proof.

Since $H \subseteq \text{CGrid}$, each vertex has distinct x, y coordinates. Name vertex i by its rank sorting by $x + \frac{y}{n}$. Denote π_i by the rank of vertex i sorted by $y + \frac{x}{n}$.

Conversely, given a permutation $\pi \in S_n$, construct vertices (i, π_i) for $i = 1, \dots, n$. This produces a subgraph of CGrid with n vertices.

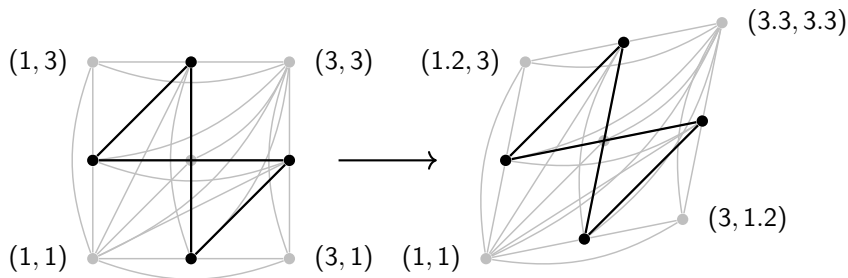
These are inverses of each other.



Example Permutation Representation

Example

The following subgraph H has $\pi(H) = (2, 1, 4, 3)$, since the leftmost vertex is 2nd lowest in y order, and so on.



The Enumeration Algorithm

Lemma

*Let H be a subgraph with $|V(H)| = O(n)$ and $|E(H)| = O(n)$.
Using a Fenwick tree, we can construct the degree sequence of H
from $\pi(H)$ in two passes of $O(n \log n)$ time.*

The Enumeration Algorithm

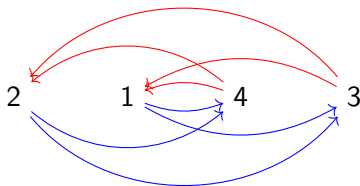
Lemma

Let H be a subgraph with $|V(H)| = O(n)$ and $|E(H)| = O(n)$. Using a Fenwick tree, we can construct the degree sequence of H from $\pi(H)$ in two passes of $O(n \log n)$ time.

In the **first pass**, construct the **backward degree**, i.e. $\forall j < i$, find $\pi_i > \pi_j$. In the **second pass**, run the algorithm on $\pi(H)$ reversed to get the **forward degree**, i.e. $\forall j > i$, find $\pi_i < \pi_j$.

The sum of both is the degree sequence.

Example



The Enumeration Algorithm cont'd

Lemma

Without even completing the *first pass*, we can identify if H , $|V(H)| = n$, has more than $c = O(n)$ edges. By setting a corresponding minimum, we can enumerate all H with $|V(H)| \in [b, c]$ with $O(n \log n)$ processing for each H .

Example



The Enumeration Algorithm cont'd

Theorem

Given an $n \in \mathbb{N}$, we can obtain a (small) set H with

- ▶ $|V(H)| = n$
- ▶ *at least the number of edges in the target Grid*
- ▶ *at most k additional edges compared to the target Grid*
- ▶ *degree sequence at least the degree sequence of target Grid*
- ▶ *(some others)*

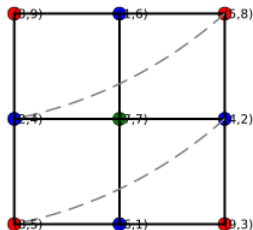
with $O(n \log n)$ processing for each H .

Recall: Which grids are subgraphs?

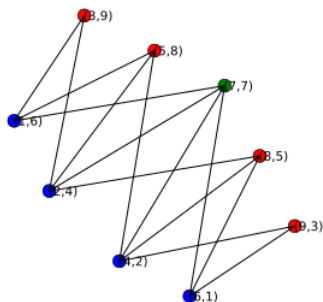
Theorem

$\text{Grid}_{k \times n}$ is not a subgraph of CGrid for $k > 2$. For $k = 3$, the minimum is 2 additional edges.

Embedding 1



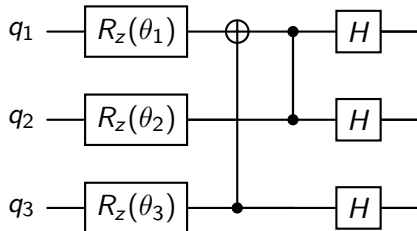
Comparability graph



The Width-3 Gadget

Definition

The previous graph gives rise to the following width-3 gadget:



Question: Is this gadget universal for MBQC?

Lie Algebra Universality Algorithm

Definition (Matrix Exponentiation)

For any $A \in \mathbb{C}^{m \times m}$,

$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!}$$

Lie Algebra Universality Algorithm

Definition (Matrix Exponentiation)

For any $A \in \mathbb{C}^{m \times m}$,

$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!}$$

Definition (Lie Algebra of $SU(2^n)$)

$$\mathfrak{su}(2^n) = \{-iH \in \mathbb{C}^{2^n \times 2^n} \mid H^\dagger = H, \operatorname{tr}(H) = 0\}$$

Lie Algebra Universality Algorithm cont'd

Theorem

A gate set $\mathcal{G}$ is universal for n qubits if it generates $SU(2^n)$, the group of $2^n \times 2^n$ unitary matrices with determinant 1.

Lie Algebra Universality Algorithm cont'd

Theorem

A gate set $\mathcal{G}$ is universal for n qubits if it generates $SU(2^n)$, the group of $2^n \times 2^n$ unitary matrices with determinant 1.

Fact

$U \in SU(2^n)$ if and only if $U = e^{iH}$ for some $H \in \mathfrak{su}(2^n)$.

Corollary

To show $\mathcal{G}$ is universal, it suffices to generate the Lie algebra.

Lie Algebra Universality Algorithm cont'd

Theorem

A gate set $\mathcal{G}$ is universal for n qubits if it generates $SU(2^n)$, the group of $2^n \times 2^n$ unitary matrices with determinant 1.

Fact

$U \in SU(2^n)$ if and only if $U = e^{iH}$ for some $H \in \mathfrak{su}(2^n)$.

Corollary

To show $\mathcal{G}$ is universal, it suffices to generate the Lie algebra.

Fact

$H \in \mathfrak{su}(2^n)$ if and only if $H = -i \sum_{P \in \mathcal{P} \setminus \{I\}} \alpha_P P$ for some $\alpha_P \in \mathbb{R}$, where $\mathcal{P}$ is the set of Pauli matrices on n qubits.

Corollary

The traceless Hermitian operators are exactly those that can be represented as a linear combination of Pauli operators (X , Y , Z).

Lie Algebra Universality Algorithm: Cliffords

Lemma

Denote by $G(\theta_1, \theta_2, \theta_3)$ the width-3 gadget. Then $G(0, 0, 0)$ is some fixed Clifford operation C and $C \in \mathcal{G}$.

Lie Algebra Universality Algorithm: Cliffords

Lemma

Denote by $G(\theta_1, \theta_2, \theta_3)$ the width-3 gadget. Then $G(0, 0, 0)$ is some fixed Clifford operation C and $C \in \mathcal{G}$.

Lemma

If $C \in \mathcal{G}$ then $C^\dagger \in \mathcal{G}$.

Lie Algebra Universality Algorithm: Cliffords

Lemma

Denote by $G(\theta_1, \theta_2, \theta_3)$ the width-3 gadget. Then $G(0, 0, 0)$ is some fixed Clifford operation C and $C \in \mathcal{G}$.

Lemma

If $C \in \mathcal{G}$ then $C^\dagger \in \mathcal{G}$.

Proof.

The Clifford group on 3 qubits is finite. Hence every element has finite order, i.e. $\exists t$ with $C^t = I$. Thus $C^\dagger = C^{t-1} \in \mathcal{G}$. □

Lie Algebra Universality Algorithm: Addition

Fact (Baker-Campbell-Hausdorff formula)

For $A, B \in \mathfrak{su}(2^n)$,

$$e^A e^B = e^{A+B+\frac{1}{2}[A,B]+\frac{1}{12}[A,[A,B]]-\frac{1}{12}[B,[A,B]]+\dots}$$

where $[A, B] = AB - BA$ is the commutator of A and B .

Fact

For $A, B \in \mathfrak{su}(2^n)$,

$$\lim_{k \rightarrow \infty} e^{A/k} e^{B/k} = e^{A+B}$$

Corollary

If we have A, B in the Lie algebra, we can generate their sum.

Lie Algebra Universality Algorithm: Commutators

Fact

For $A, B \in \mathfrak{su}(2^n)$,

$$\lim_{k \rightarrow \infty} e^{B/k} e^{A/k} e^{-B/k} e^{-A/k} = e^{[A,B]}.$$

Corollary

If we have A, B in the Lie algebra, we can generate their commutator $[A, B]$.

Lie Algebra Universality Algorithm: Operations

Strategy: Start with some elements in the Lie algebra and generate more using:

- ▶ Conjugation by C
- ▶ Addition
- ▶ Commutators

Lie Algebra Universality Algorithm: Operations

Strategy: Start with some elements in the Lie algebra and generate more using:

- ▶ Conjugation by C
- ▶ Addition
- ▶ Commutators

Fact

We cannot generate all $|\{I, X, Y, Z\}|^3 = 64$ Pauli operators on 3 qubits this way.

Lie Algebra Universality Algorithm: Operations

Strategy: Start with some elements in the Lie algebra and generate more using:

- ▶ Conjugation by C
- ▶ Addition
- ▶ Commutators

Fact

We cannot generate all $|\{I, X, Y, Z\}|^3 = 64$ Pauli operators on 3 qubits this way.

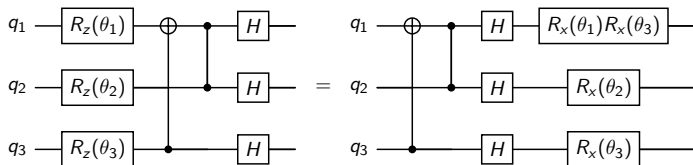
Proof.

Brute force using a computer program.



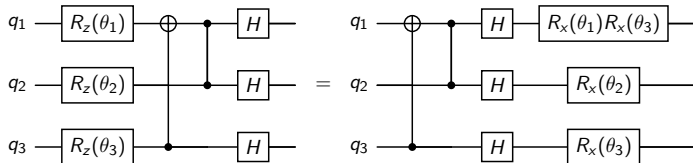
Lie Algebra Universality Algorithm: Visual Intuition

The *CNOT* “forbids us” from generating $R_x(\theta_1)$ alone on q_1 .



Lie Algebra Universality Algorithm: Visual Intuition

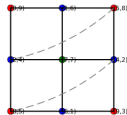
The *CNOT* “forbids us” from generating $R_x(\theta_1)$ alone on q_1 .



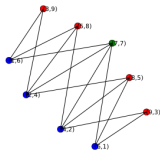
Idea: “Stitch” this with another gadget to “get” $R_x(\theta_1)$ on q_1 .

Recall: Many width-3 gadgets.

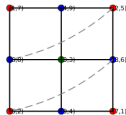
Embedding 1



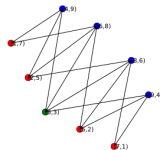
Comparability graph



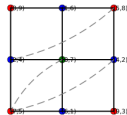
Embedding 1



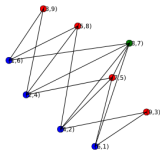
Comparability graph



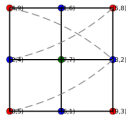
Embedding 1



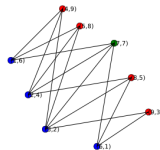
Comparability graph



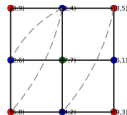
Embedding 1



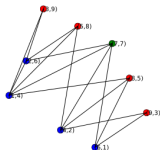
Comparability graph



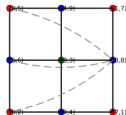
Embedding 1



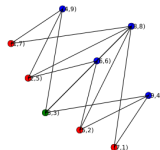
Comparability graph



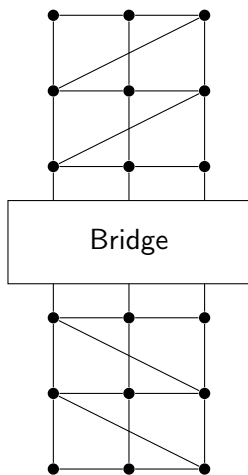
Embedding 1



Comparability graph



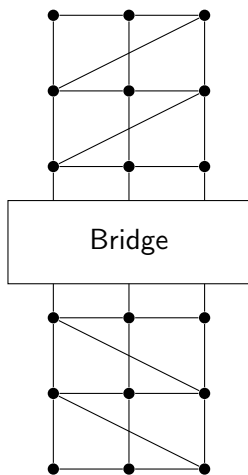
Bridging Gadgets: Informal Visual Definition



Definition (Informal)

A width-3 bridge of size $b \in \mathbb{N} \cup \{0\}$ adds $3b$ new vertices between two width-3 gadgets. The bridge has some $\text{Grid}_{3 \times b}$ as a subgrid.

Bridging Gadgets: Informal Visual Definition

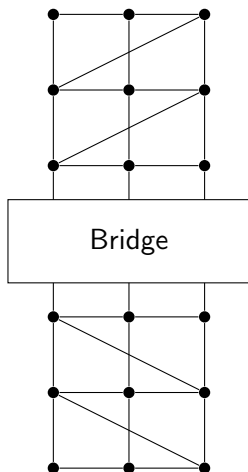


Definition (Informal)

A width-3 bridge of size $b \in \mathbb{N} \cup \{0\}$ adds $3b$ new vertices between two width-3 gadgets. The bridge has some $\text{Grid}_{3 \times b}$ as a subgrid.

The bridge can have any number of additional edges; set $\theta = 0$ to ignore them.

Bridging Gadgets: Informal Visual Definition



Definition (Informal)

A width-3 bridge of size $b \in \mathbb{N} \cup \{0\}$ adds $3b$ new vertices between two width-3 gadgets. The bridge has some $\text{Grid}_{3 \times b}$ as a subgrid.

The bridge can have any number of additional edges; set $\theta = 0$ to ignore them.

Question: Given width-3 gadgets A and B , how can we construct a width-3 bridge of size b or prove that it is impossible?

Linear Programming Bridging Algorithm

Definition (Informal)

Given width-3 gadgets A and B , encode each as inequalities on a linear program.

Example

Suppose we are dealing with width-2 gadgets and A has $\pi(A) = (2, 1, 4, 3)$. We add conditions:

$$x_1 < x_2 < x_3 < x_4$$

$$y_2 < y_1 < y_4 < y_3.$$

Linear Programming Bridging Algorithm cont'd

Lemma (Informal)

We can add additional inequalities to a linear program encoding A to also encode B and any bridge vertices.

Theorem

Fix a bridge size b . If the corresponding linear program is infeasible, then there is no bridge of size b between A and B .

Linear Programming Bridging Algorithm cont'd

Lemma (Informal)

We can add additional inequalities to a linear program encoding A to also encode B and any bridge vertices.

Theorem

Fix a bridge size b . If the corresponding linear program is infeasible, then there is no bridge of size b between A and B .

Fact

There exists no bridge of size $b < 4$ between any two width-3 gadgets A, B , where each gadget has at most 3 additional edges compared to $\text{Grid}_{3 \times 3}$.

Linear Programming Bridging Algorithm cont'd

Lemma (Informal)

We can add additional inequalities to a linear program encoding A to also encode B and any bridge vertices.

Theorem

Fix a bridge size b . If the corresponding linear program is infeasible, then there is no bridge of size b between A and B .

Fact

There exists no bridge of size $b < 4$ between any two width-3 gadgets A, B , where each gadget has at most 3 additional edges compared to $\text{Grid}_{3 \times 3}$.

Proof.

Brute force every linear program encoding of ordered pairs of gadgets A, B with at most 3 additional edges.



Resolution?

Conjecture

CGrid is not a universal resource for MBQC.

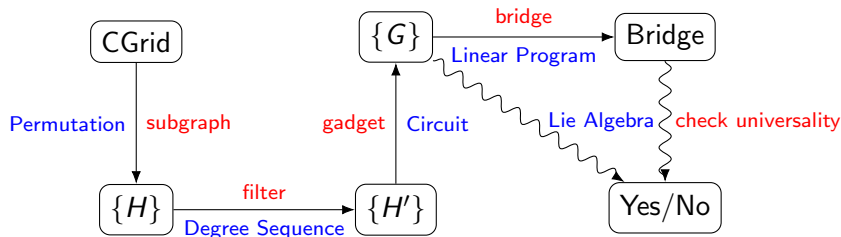
Resolution?

Conjecture

CGrid is not a universal resource for MBQC.

Justification

Considering gadgets with more additional edges is unlikely to help get $R_z(\theta_1)$ on q_1 .



Why Care about Comparability Grids?

Theorem (Maarten Van den Nest et al, 2006)

Let Ψ be a family of resource states, and $|\phi\rangle, |\phi'\rangle$ arbitrary states with the same number of qubits. Let $E : |\phi\rangle \rightarrow \mathbb{R}$ be a function that satisfies the following properties:

1. $E(|\phi\rangle) \geq E(|\phi'\rangle)$ whenever $|\phi\rangle \geq_{\text{LOCC}} |\phi'\rangle$;
2. $\sup_{|\phi\rangle} E(|\phi\rangle) > \sup_{|\psi\rangle \in \Psi} E(|\psi\rangle)$.

Then Ψ cannot be a universal resource.

Why Care about Comparability Grids?

Theorem (Maarten Van den Nest et al, 2006)

Let Ψ be a family of resource states, and $|\phi\rangle, |\phi'\rangle$ arbitrary states with the same number of qubits. Let $E : |\phi\rangle \rightarrow \mathbb{R}$ be a function that satisfies the following properties:

- 1. $E(|\phi\rangle) \geq E(|\phi'\rangle)$ whenever $|\phi\rangle \geq_{\text{LOCC}} |\phi'\rangle$;*
- 2. $\sup_{|\phi\rangle} E(|\phi\rangle) > \sup_{|\psi\rangle \in \Psi} E(|\psi\rangle)$.*

Then Ψ cannot be a universal resource.

Theorem

CGrids have unbounded rank width (and so do grid graphs).

Why Care about Comparability Grids?

Theorem (Maarten Van den Nest et al, 2006)

Let Ψ be a family of resource states, and $|\phi\rangle, |\phi'\rangle$ arbitrary states with the same number of qubits. Let $E : |\phi\rangle \rightarrow \mathbb{R}$ be a function that satisfies the following properties:

- 1. $E(|\phi\rangle) \geq E(|\phi'\rangle)$ whenever $|\phi\rangle \geq_{\text{LOCC}} |\phi'\rangle$;*
- 2. $\sup_{|\phi\rangle} E(|\phi\rangle) > \sup_{|\psi\rangle \in \Psi} E(|\psi\rangle)$.*

Then Ψ cannot be a universal resource.

Theorem

CGrids have unbounded rank width (and so do grid graphs).

Corollary

If CGrid is not a universal resource, then unbounded rank width is not sufficient for universality.